



$$\sum \epsilon_i = 0$$

$$\sum \epsilon_i^2 = \text{Obj. fn.}$$

Minimize the $\sum \epsilon_i^2$

$$\sum |\epsilon_i|$$

$$Y'X\beta = 1 \times n \times n \times 1 \times 1 \times n \times 1 \times 1 = 1$$

How to get Least Squares Estimate of the Model Parameters?

Loss fn.

$$L = E'E = (Y - X\beta)'(Y - X\beta)$$

$$= (Y' - \beta'X')(Y - X\beta) \quad \because (AB)' = B'A'$$

$$L = Y'Y - Y'X\beta - \beta'X'Y + \beta'X'X\beta \quad (X\beta)' = \beta'X'$$

$$L = Y'Y - 2\beta'X'Y + \beta'X'X\beta$$

$$\frac{\partial L}{\partial \beta} = 0 - 2X'Y + 2X'X\beta = 0$$

$$Y'X\beta$$

$$1 \times n \times n \times 1 \times 1 \times n \times 1 = 1 \times 1$$

scalar

$$\beta'X'Y$$

$$1 \times 1 \times n \times n \times n \times 1 \times 1 = 1 \times 1$$

scalar

$$X'X\beta = X'Y$$

$$\hat{\beta} = (X'X)^{-1}(X'Y)$$

Least Squares predictor

$$(X'X)^{-1}$$

Carries a lot of importance

$$(\epsilon\epsilon')$$

$$[\epsilon_1 \quad \epsilon_2 \quad \dots \quad \epsilon_n] \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} = \epsilon_1^2 + \epsilon_2^2 + \dots + \epsilon_n^2 = \sum_{j=1}^n \epsilon_j^2$$

$$e'e = [Y - X\hat{\beta}]' [Y - X\hat{\beta}]$$

$$= [Y' - \hat{\beta}'X'] [Y - X\hat{\beta}] = Y'Y - \hat{\beta}'X'Y$$

$$\hat{\beta} = (X'X)^{-1}X'Y$$

$$AA' = I$$

$$e'e = Y'Y - Y'X\hat{\beta} - \hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta}$$

$$= Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'X(X'X)^{-1}X'Y$$

$$Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'Y \rightarrow I$$

$$e'e = Y'Y - \hat{\beta}'X'Y$$

$$\text{Residual SSQ} = \text{Total SSQ} - \text{Regression SSQ}$$

Very 'X'

$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n y_i^2$$

$$Y = X\beta + \epsilon$$

true observation

$$\hat{Y} = X\hat{\beta}$$

No error term

Least Sq. Estimation stores

Purity 20 ± 0.5

$$\beta = 1.3 = 5$$

$$\hat{\beta} = 4.2 \pm 1.0$$

$$\frac{3.2 - 4.8}{3.2 - 5.2}$$